

UNIVERSITY OF MILANO-BICOCCA

School of Economics and Statistics

Master's Degree Program in

STATISTICAL AND ECONOMIC SCIENCES



**A GENERALIZATION OF THE CENTERED
ERDŐS-RÉNYI DISTRIBUTION FOR NETWORK DATA
ANALYSIS**

Supervisor: Prof. Bernardo Nipoti

Co-supervisor: Prof. Tommaso Rigon

Master's Thesis by:

Sara Capozio

Student ID No. 853600

Academic Year 2023/2024

Contents

Introduction	1
1 Generalized Centered Erdős–Rényi Model	3
1.1 Preliminary Aspects	3
1.2 Introduction to the Generalized Erdős–Rényi Model	5
2 Properties of the GCER Model	10
2.1 Properties Related to the Concentration Parameter	10
2.2 Properties Related to the Scale Parameter	16
3 Bayesian Inference for the GCER Model	20
4 Bayesian Inference for the CER Model with a GCER Prior	22
4.1 GCER Distribution as a Conjugate Prior in a CER Model	22
4.2 Unknown Scale Parameter α	25
4.2.1 Prior with Dependent Components	25
4.2.2 Prior with Independent Components	28
5 Discriminant Analysis as an Application Example	30
5.1 Homoscedasticity Assumption	31
5.2 Heteroscedasticity Assumption	32
6 Simulation Study	34
7 Analysis of the Human Brain Networks Data	39
Conclusions	49
Bibliography	51

Introduction

Multiple networks represent a type of data that can be used to describe interactions or connections among the elements of a system of varying complexity. They consist of a collection of graphs characterized by the same number of nodes, which are related to one another through interactions. The distinctive features of each graph are determined by the connections among its nodes, namely by its edge structure.

Multiple networks find applications in several research fields, particularly in neuroscience. In this context, a population of networks can be used to represent interactions among different brain regions across multiple individuals, whose analysis may help identify the presence of neurological disorders.

Several probabilistic models have been introduced to model this type of data, including the Erdős–Rényi (ER) model, which is based on the assumption that each connection between two distinct nodes is independent and identically distributed according to a Bernoulli distribution. In the work by Lunagómez et al. (2021), an extension of the ER model, namely the centered Erdős–Rényi (CER) model, was introduced. Compared with the ER model, the CER model is more flexible, as it relaxes the assumption that connections between pairs of distinct nodes are identically distributed. The probability of observing an edge between two nodes is no longer the same for all pairs, but depends on two parameters: a location parameter and a scale parameter. The location parameter coincides with the mode of the distribution and represents a measure of centrality, whereas the scale parameter measures the degree of concentration of the observations around the location parameter.

Although the CER model is characterized by an easily interpretable analytical form, its probabilistic structure is relatively rigid. This may represent a limitation when analyzing interaction structures complex that require more flexible approaches.

This thesis introduces a generalization of the centered Erdős–Rényi model, referred to as the generalized centered Erdős–Rényi model, abbreviated as GCER. This new model has the same parametric structure as the CER model, but, unlike the latter, the location parameter is represented by a weighted graph defined on a larger space than the space of binary graphs.

The aim of this thesis is to provide a detailed analysis of the new GCER probabilistic model by investigating its main properties, inferential strategies, and potential applications.

The first part of the thesis focuses on the definition of the model and its main properties. In

this section, the relationship between the CER and GCER models is examined, highlighting how some characteristic properties of the more restrictive model can be generalized to the more flexible setting. This analysis makes it possible to emphasize not only the similarities between the two approaches but also the advantages of using the GCER model.

The discussion then proceeds with an analysis of the Bayesian inferential strategies used to estimate the parameters of the generalized centered Erdős–Rényi model, highlighting the analytical and computational limitations of this approach.

The issues encountered when using the GCER model as a likelihood function do not arise when this model is used as a prior distribution for the CER model. Indeed, the GCER model is conjugate with respect to the CER model, provided that the scale parameter is known. This property allows the posterior distribution of the location parameter to be easily derived analytically, thereby simplifying inference. However, the conjugacy property does not hold when the scale parameter is unknown; for this reason, simulation-based computational techniques, such as Gibbs sampling, are required.

The second part of the thesis focuses on a simulation study aimed at illustrating the flexibility of the GCER model compared with the CER model and on the application of the inferential methodologies introduced above to the real-world ‘Human Brain Networks’ dataset, with the aim of modelling the structural characteristics of the brain networks of thirty patients.

The thesis concludes with an application based on discriminant analysis, which leads to the formulation of the corresponding decision rule and to the evaluation of the classification performance of the CER model under the assumption of heteroscedastic observations.

1. Generalized Centered Erdős–Rényi Model

1.1. Preliminary Aspects

A simple undirected labeled binary graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ consists of a set of labeled nodes $\mathcal{V} = \{1, \dots, N\}$ and a set of edges $\mathcal{E} \subseteq \{\mathcal{E} \subset \mathcal{V} : |\mathcal{E}| = 2\}$. The total number of nodes is therefore denoted by $N = |\mathcal{V}|$, while

$$\mathcal{G}_{[N]} := \{\mathcal{G} = (\mathcal{V}, \mathcal{E}) : \mathcal{V} = \{1, \dots, N\}\},$$

represents the set of simple undirected labeled binary graphs with nodes $\mathcal{V}$, also referred to as the graph space. When considering only simple graphs, the total number of possible graphs is given by $|\mathcal{G}_{[N]}| = 2^M$, where $M = N(N-1)/2$ represents the maximum number of possible connections for a graph with N nodes. Each graph $\mathcal{G}$ can be represented by an adjacency matrix $A_{\mathcal{G}}$ of dimension $N \times N$ such that:

$$A_{\mathcal{G}}[i, j] = \begin{cases} 1 & \text{if } [i, j] \in \mathcal{E} \\ 0 & \text{otherwise} \end{cases}$$

where $A_{\mathcal{G}}[i, j]$ represents the entry at position $[i, j]$ of the adjacency matrix $A_{\mathcal{G}}$ associated with the graph $\mathcal{G}$. Since undirected graphs are considered, the adjacency matrix $A_{\mathcal{G}}$ is symmetric.

In order to define the basic probabilistic model on $\mathcal{G}_{[N]}$, which forms the foundation of the present work, it is useful to introduce the concept of Hamming distance. The Hamming distance between two graphs, defined as:

$$d_H(\mathcal{G}_1, \mathcal{G}_2) = \sum_{i=1}^{N-1} \sum_{j=i+1}^N |A_{\mathcal{G}_1}[i, j] - A_{\mathcal{G}_2}[i, j]|,$$

represents a measure of similarity defined on $\mathcal{G}_{[N]}$ that corresponds to the number of connections that differ between the two graphs $\mathcal{G}_1$ and $\mathcal{G}_2$. This quantity is useful for quantifying how much two graphs differ in terms of their connection structures. The CER model is a probabilistic model defined in terms of the Hamming distance. The CER model is parametrized by a location parameter $\mathcal{G}^m$ defined on $\mathcal{G}_{[N]}$, which coincides with the mode of the distribution, and a scale parameter α that measures the variability of the graphs around the centrality parameter $\mathcal{G}^m$. To define the location parameter, the notion of the Fréchet mean is used, which is a generalization of the concept of mean to non-Euclidean spaces. In the CER distribution, the Fréchet mean $\mathcal{G}^m$ and the mode coincide. The model is based on the assumption that each edge is independently, but not identically, distributed according to a Bernoulli distribution. The probability of an edge between a pair of nodes is equal to α or $1 - \alpha$ depending on whether the edge connects the same pair of nodes as in the central graph $\mathcal{G}^m$. Formally, the model can be defined by specifying the probability of observing an edge between nodes $i \neq j$ as:

$$\Pr(A_{\mathcal{G}}[i, j] = A_{\mathcal{G}^m}[i, j]) = 1 - \alpha.$$

Expressing the model in global form yields a discrete distribution p_{CER} on $\mathcal{G}_{[N]}$ defined as:

$$p_{\text{CER}}(\mathcal{G}; \mathcal{G}^m, \alpha) = \alpha^{d_{\text{H}}(\mathcal{G}, \mathcal{G}^m)} (1 - \alpha)^{M - d_{\text{H}}(\mathcal{G}, \mathcal{G}^m)}. \quad (1.1)$$

To indicate that a random graph $\mathcal{G}$ is distributed according to the distribution in 1.1, the notation $\mathcal{G} \sim \text{CER}(\mathcal{G}^m, \alpha)$ is used.

In the following sections, whenever necessary, the more compact notation $p_{\text{CER}}(\cdot)$ will be used in place of $p_{\text{CER}}(\cdot; \mathcal{G}^m, \alpha)$.

Although the distribution (1.1) is defined for every $\alpha \in (0, 1)$, the domain of the scale parameter is restricted to the interval $(0, 1/2)$ in order to ensure unimodality of the distribution. From formula (1.1), it follows that graphs that are more concentrated around the mode $\mathcal{G}^m$ have a higher probability of being observed. In particular, if $d_{\text{H}}(\mathcal{G}_1, \mathcal{G}^m) > d_{\text{H}}(\mathcal{G}_2, \mathcal{G}^m)$, then $p_{\text{CER}}(\mathcal{G}_2) > p_{\text{CER}}(\mathcal{G}_1)$.

1.2. Introduction to the Generalized Erdős–Rényi Model

The CER model can be viewed as a generalization of the simpler Erdős–Rényi model, in which each connection is assumed to be independently and identically distributed according to a Bernoulli distribution. In the CER model, the success probability is a function of a location parameter and a scale parameter. In this case, the assumption of identically distributed edges no longer holds, since the probability of observing an edge depends on the value taken by $\mathcal{G}^m$ for the corresponding pair of nodes.

This thesis considers a further generalization of the CER model through the introduction of a new location measure represented by a simple undirected labeled weighted graph W . This new parameter is defined on a larger space than the space of binary graphs, denoted by $\mathcal{W}_{[N]}$. The space $\mathcal{W}_{[N]}$ is formally defined as:

$$\mathcal{W}_{[N]} = \{W \in [0, 1]^{N \times N} : W[i, j] = W[j, i] \text{ and } W[i, i] = 0 \text{ for every } i, j \in \{1, \dots, N\}\}$$

and represents the set of all simple undirected labeled weighted graphs with N nodes, where each element $W[i, j]$ of the matrix W represents the weight of the edge between nodes i and j .

Definition 1.1. A random graph $\mathcal{G} \in \mathcal{G}_{[N]}$ follows a generalized centered Erdős–Rényi (GCER) distribution if for every $i \neq j$,

$$\Pr(\mathcal{G}[i, j] = 1; W, \alpha) = \left(1 + \left(\frac{\alpha}{1 - \alpha}\right)^{2W[i, j] - 1}\right)^{-1}. \quad (1.2)$$

To indicate that a random graph $\mathcal{G}$ is distributed according to this model, the notation $\mathcal{G} \sim \text{GCER}(W, \alpha)$ is used.

As in the case of the CER distribution, each edge is independently, but not identically, distributed according to a Bernoulli distribution with success probability defined in (1.2), which can be formally expressed as:

$$\mathcal{G}[i, j] \sim \text{Bernoulli} \left(\left(1 + \left(\frac{\alpha}{1 - \alpha} \right)^{2W[i, j] - 1} \right)^{-1} \right).$$

The expected value of the edge between the pair of nodes i and j coincides with the probability of observing a connection between them. Therefore:

$$\mathbb{E}[\mathcal{G}[i, j]] = \left(1 + \left(\frac{\alpha}{1 - \alpha} \right)^{2W[i, j] - 1} \right)^{-1}. \quad (1.3)$$

The GCER probabilistic model is more flexible than the CER model, in which the probability of observing a connection between two nodes can take only two values, depending on the presence of an edge connecting the same nodes in $\mathcal{G}^m$. In the new model, the introduction of a weighted graph as the location matrix overcomes the rigid structure of the CER model, allowing greater flexibility in modelling connection probabilities. This new distribution can be viewed as a compromise between the CER model and the more general model in which each edge is independently distributed with a success probability defined by a parameter specific to each pair of nodes. Although the latter represents the highest level of flexibility, it has the disadvantage of reducing parameter interpretability and losing the convenient distinction between the location parameter and the scale parameter that characterizes the CER model. In contrast, the GCER model preserves this property through its parametric structure. Indeed, in the GCER model, the success probability is a function of two distinct parameters, each with a specific interpretation: a location parameter W and a scale parameter α .

The parameter W is a location parameter defined on the space $\mathcal{W}_{[N]}$, represented by a symmetric $N \times N$ matrix of weights. From equation (1.3), it is possible to express $W[i, j]$ as a function of the parameter α and the expected value $\mathbb{E}[i, j]$, obtaining:

$$W[i, j] = \frac{1}{2} \left(\frac{\log \left(\frac{1 - \mathbb{E}[\mathcal{G}[i, j]]}{\mathbb{E}[\mathcal{G}[i, j]]} \right)}{\log \left(\frac{\alpha}{1 - \alpha} \right)} + 1 \right).$$

Each element $W[i, j] \in [0, 1]$ measures the strength of the connection between the pair of

nodes i and j , affecting the probability of observing an edge between them. Figure 1.1 shows that the closer $W[i, j]$ is to its maximum value, the stronger the connection between the pair of nodes i and j and, consequently, the higher the probability of observing an edge between them. Conversely, the closer $W[i, j]$ is to zero, the weaker the connection.

The parameter α , as in the case of the CER distribution, is a scale parameter that takes values in the interval $(0, 1/2)$ and quantifies the variability of the graph around the location parameter W . As shown in Figure 1.1 for low values of α , the concentration around W is highest; conversely, for high values of α , the probability of observing an edge between nodes i and j approaches $1/2$ for any value of $W[i, j]$, indicating greater variability in the graph structure.

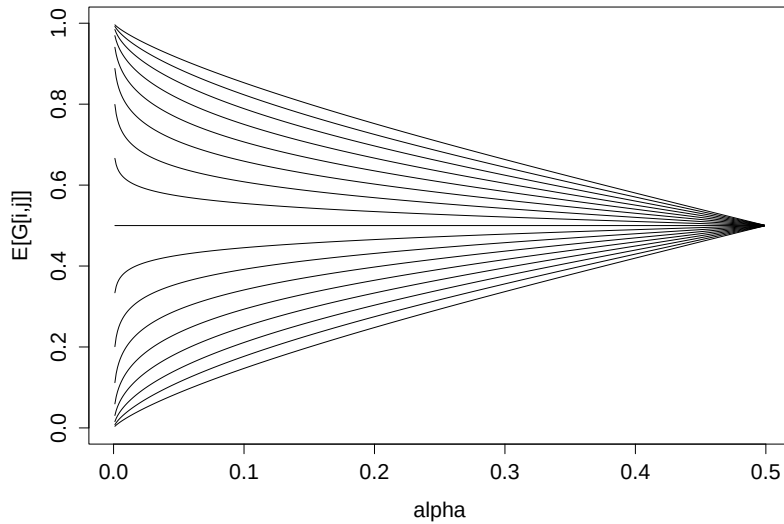


Figure 1.1: Expected value $\mathbb{E}[\mathcal{G}[i, j]]$ as a function of $\alpha \in (0, 1/2)$ for different values of $W[i, j] \in \{0.1, \dots, 0.9\}$ (from bottom to top).

This new distribution is referred to as the generalized centered Erdős–Rényi distribution because it includes the CER distribution as a special case. Indeed, setting $W = \mathcal{G}^m \in \mathcal{G}_{[N]}$:

$$\Pr(\mathcal{G}[i, j] = 1; \mathcal{G}^m, \alpha) = \begin{cases} 1 - \alpha & \text{if } \mathcal{G}^m[i, j] = 1 \\ \alpha & \text{if } \mathcal{G}^m[i, j] = 0 \end{cases}$$

If the location matrix W belongs to the space $\mathcal{G}_{[N]}$, then the GCER distribution reduces to the CER distribution and the location parameter W therefore coincides with the mode of the

distribution.

To better understand the behavior of the probability of observing a connection between the pair of nodes i and j as the parameters α and W vary, as defined in equation (1.2), we analyze how the distribution of a random graph $\mathcal{G} \sim \text{GCER}(W, \alpha)$ behaves in the limiting cases:

- For $\alpha \rightarrow 0$ and $W[i, j] > 1/2$, $\Pr(\mathcal{G}[i, j] = 1) = 1$. The presence of an edge between nodes i and j of the graph $\mathcal{G}$, as α tends to zero, is certain if $W[i, j] > 1/2$. Conversely, for very small values of α and weights below $1/2$, the probability of observing a connection between the pair of nodes i and j is zero. In these two scenarios, the probability distribution is degenerate since, depending on the values taken by $W[i, j]$, the probability of observing or not observing an edge between the pair of nodes i and j reaches its limiting value.
- For $\alpha \in (0, 1/2)$, $W[i, j] = 1/2$ and for $\alpha \rightarrow 1/2$, $W[i, j] \in [0, 1]$, $\Pr(\mathcal{G}[i, j] = 1) = 1/2$. These two scenarios represent situations of maximum uncertainty in which $\mathcal{G}[i, j] \sim \text{Bernoulli}(1/2)$. In these cases, the GCER distribution reduces to the classical Erdős–Rényi distribution.

For $g \in \{0, 1\}$, the GCER distribution can be written in expanded form as:

$$\begin{aligned}
\Pr(\mathcal{G}[i, j] = g; W, \alpha) &= \left[1 + \left(\frac{\alpha}{1-\alpha} \right)^{2W[i, j]-1} \right]^{-g} \left[1 - \left(1 + \left(\frac{\alpha}{1-\alpha} \right)^{2W[i, j]-1} \right)^{-1} \right]^{1-g} \\
&= \left[1 + \left(\frac{\alpha}{1-\alpha} \right)^{2W[i, j]-1} \right]^{-g} \left[\frac{\left(\frac{\alpha}{1-\alpha} \right)^{2W[i, j]-1}}{1 + \left(\frac{\alpha}{1-\alpha} \right)^{2W[i, j]-1}} \right]^{1-g} \\
&= \left(\frac{\alpha}{1-\alpha} \right)^{g(1-2W[i, j])} \left[\frac{\left(\frac{\alpha}{1-\alpha} \right)^{2W[i, j]-1}}{1 + \left(\frac{\alpha}{1-\alpha} \right)^{2W[i, j]-1}} \right].
\end{aligned}$$

This leads to an equivalent definition of the GCER distribution to that given in Definition 1.1.

Definition 1.2. A random graph $\mathcal{G} \in \mathcal{G}_{[N]}$ follows a generalized centered Erdős–Rényi (GCER) distribution if

$$p(\mathcal{G}; W, \alpha) = \left(\frac{\alpha}{1 - \alpha} \right)^{\sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}[i,j](1-2W[i,j])} \frac{1}{\prod_{i=1}^{N-1} \prod_{j=i+1}^N \left(1 + \left(\frac{\alpha}{1-\alpha} \right)^{1-2W[i,j]} \right)}$$

or equivalently, up to a normalizing constant,

$$p(\mathcal{G}; W, \alpha) \propto \left(\frac{\alpha}{1 - \alpha} \right)^{\sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}[i,j](1-2W[i,j])}.$$

It is important to emphasize that, although the model is well defined for $\alpha \in (0, 1)$, the domain of the parameter is restricted to the interval $(0, 1/2)$ in order to guarantee certain properties of the distribution, which will be stated in Chapter 2, and, above all, to preserve the interpretability of the model. Indeed, if values of $\alpha \in (1/2, 1)$ were considered, graph configurations far from the centrality measure W would be favored.

2. Properties of the GCER Model

This section presents the properties that characterize the GCER probabilistic model and its relationship with the CER model. The analysis of these properties allows us to define the role of the location parameter W and the scale parameter α . First, the parameter W is analyzed, highlighting its relationship with the location parameter $\mathcal{G}^m$ of the CER model and its role in representing a measure of centrality for the distribution. Subsequently, the analysis focuses on the scale parameter α , demonstrating its ability to quantify the variability of the distribution.

2.1. Properties Related to the Concentration Parameter

In order to study the role of the parameter W in the GCER model, it is necessary to introduce the following notion:

Definition 2.1. Let Y be a random variable defined on the space $\mathcal{Y}$ and let $d_H(\cdot, \cdot)$ be the Hamming distance defined on $\mathcal{Y}$. The solution to the minimization problem:

$$\phi^m = \underset{\phi \in \mathcal{Y}}{\operatorname{argmin}} \mathbb{E}[d_H^2(Y, \phi)]$$

is called the Fréchet mean.

Proposition 2.1. : Let $\mathcal{G} \sim \text{GCER}(W, \alpha)$ be a random graph defined on $\mathcal{G}_{[N]}$, then the Fréchet mean of the distribution can be interpreted as a discretization of the weighted graph W :

$$\mathcal{G}^m = \lfloor 2W \rfloor ,$$

where $\lfloor 2W \rfloor$ is obtained by taking the integer part of each element $2W[i, j]$ of the matrix

$2W$.

Proof. Let Y be a random graph defined on $\mathcal{G}_{[N]}$, the Fréchet mean is obtained as the solution to the minimization problem:

$$\mathcal{G}^m = \operatorname{argmin}_{\mathcal{G} \in \mathcal{G}_{[N]}} \mathbb{E}[d_H^2(Y, \mathcal{G})],$$

where the squared Hamming distance is defined as:

$$d_H^2(Y, \mathcal{G}) = \sum_{i=1}^{N-1} \sum_{j=i+1}^N (|Y[i, j] - \mathcal{G}[i, j]|)^2 = \sum_{i=1}^{N-1} \sum_{j=i+1}^N |Y[i, j] - \mathcal{G}[i, j]|.$$

The expected value of the squared Hamming distance can be expressed as follows:

$$\mathbb{E}[d_H^2(Y, \mathcal{G})] = \sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathbb{E}[|Y[i, j] - \mathcal{G}[i, j]|].$$

Under the assumption that an edge exists between the pair of nodes i and j , namely $\mathcal{G}[i, j] = 1$, the expected squared error is given by $\mathbb{E}[d_H^2(Y[i, j], 1)] = 1 - p_{i,j}$. Conversely, in the absence of a connection between the pair of nodes i and j , the mean squared error is $\mathbb{E}[d_H^2(Y[i, j], 0)] = p_{i,j}$. It can be observed that if $p_{i,j} < 1/2$, the mean squared error is minimized by choosing $\mathcal{G}[i, j] = 0$. Conversely, if $p_{i,j} > 1/2$, the error is minimized when $\mathcal{G}[i, j] = 1$. Therefore, the Fréchet mean $\mathcal{G}^m$ can be rewritten as:

$$\mathcal{G}^m[i, j] = \begin{cases} 1 & \text{if } p_{i,j} > 1/2 \\ 0 & \text{if } p_{i,j} < 1/2 \end{cases}$$

where

$$p_{i,j} = \Pr(\mathcal{G}[i, j] = 1) = \left(1 + \left(\frac{\alpha}{1 - \alpha} \right)^{(2W[i,j]-1)} \right)^{-1}.$$

The probability of observing an edge between the pair of nodes i and j can be explicitly rewritten by considering the condition $p_{i,j} > 1/2$, as follows:

$$p_{i,j} > 1/2 \Leftrightarrow \left(1 + \left(\frac{\alpha}{1-\alpha}\right)^{(2W[i,j]-1)}\right)^{-1} > 1/2 \Leftrightarrow \left(\frac{\alpha}{1-\alpha}\right)^{(2W[i,j]-1)} < 1$$

For $0 < \alpha < 1/2$, the condition above is satisfied if $W[i, j] > 1/2$. The Fréchet mean can therefore be rewritten as:

$$\mathcal{G}^m[i, j] = \begin{cases} 1 & \text{if } W_{i,j} > 1/2 \\ 0 & \text{if } W_{i,j} < 1/2 \end{cases}$$

or, in compact form, as:

$$\mathcal{G}^m[i, j] = \lfloor 2W[i, j] \rfloor.$$

□

Proposition 2.2. The mode and the Fréchet mean coincide for the G-CER model.

Proof. The proof of Proposition 2.2 is divided into two parts. The first part of the proof coincides with the first part of Proposition 3.6 in the paper by Lunagómez et al. (2021) .

Part 1: Let a and b be two vectors with s elements; furthermore, let w be such that $w_i > 0$ and $\sum_i w_i = 1$, and let a be such that $a_i \geq 0$. Let σ be a permutation of $\{1, 2, \dots, s\}$ such that $w_{\sigma[i]} \geq w_{\sigma[j]}$ for every pair $[i, j]$ with $i < j$. Let τ be a permutation of $\{1, 2, \dots, s\}$ such that $\tau_{\sigma[i]} \leq \tau_{\sigma[j]}$ for every pair $[i, j]$ with $i < j$. Therefore, τ satisfies:

$$\tau = \operatorname{argmin}_{\phi \in \operatorname{Sym}(s)} \sum a_{\phi[i]} \times w_{\sigma[i]}$$

where $\operatorname{Sym}(s)$ denotes the set of all permutations of the s indices. This statement is proved by induction:

for $s = 2$: consider the case $a_{\tau[1]} < a_{\tau[2]}$ and $w_{\tau[1]} > w_{\tau[2]}$. Since the elements of the vector w are non-negative and sum to one, it follows that $w_{\sigma[1]} > 1/2 > w_{\sigma[2]}$. Then the quantity

$$a_{\tau[1]} \times w_{\sigma[1]} + a_{\tau[2]} \times w_{\sigma[2]}$$

is closer to $a_{\tau[1]}$ than to $a_{\tau[2]}$. If the indices of a_{τ} are permuted to obtain a new vector $a_{\tau'}$, then

$$a_{\tau'[1]} \times w_{\sigma[1]} + a_{\tau'[2]} \times w_{\sigma[2]}$$

is closer to $a_{\tau[2]}$ than to $a_{\tau[1]}$. Since $a_{\tau[1]} < a_{\tau[2]}$, the condition is satisfied

for $s = k$: assume that the result holds for $s = k$

for $s = k + 1$: consider two cases, defined in terms of the existence of fixed points of τ .

Case 1. At least one element of a_{τ} remains fixed. Without loss of generality, it can be assumed that the element of a_{τ} that remains unchanged is the $(k + 1)$ -th element. It is possible to renormalize the first k elements of w_{σ} by defining:

$$w'_{\sigma[i]} = \frac{w_{\sigma[i]}}{1 - w_{\sigma[k+1]}}.$$

Since $w'_{\sigma[i]} > w'_{\sigma[j]}$ for every $1 \leq i < j \leq k$, the induction hypothesis can be applied to the first k entries of a_{τ} , $a_{\tau'}$ and $w'_{\sigma[i]}$ to obtain:

$$\sum_{i=1}^k a_{\tau[i]} \times w'_{\sigma[i]} \leq \sum_{i=1}^k a_{\tau'[i]} \times w'_{\sigma[i]} \quad (2.1)$$

Equation (2.1) holds for any permutation τ' that leaves the element $k + 1$ unchanged with respect to τ . Therefore:

$$\sum_{i=1}^k a_{\tau[i]} \times w_{\sigma[i]} \leq \sum_{i=1}^k a_{\tau'[i]} \times w_{\sigma[i]},$$

since it is sufficient to multiply $w'_{\sigma[i]}$ by the positive constant $1 - w_{\sigma[k+1]}$. Assuming $a_{\tau[k+1]} = a_{\tau'[k+1]}$, it follows that

$$\sum_{i=1}^{k+1} a_{\tau[i]} \times w_{\sigma[i]} \leq \sum_{i=1}^{k+1} a_{\tau'[i]} \times w_{\sigma[i]}.$$

Therefore, τ minimizes $a_{\tau[i]} \times w_{\sigma[i]}$ over all permutations that leave at least one element unchanged with respect to τ .

Case 2: Consider now the case in which no element of a_{τ} remains unchanged under a new ordering τ' . Let a_{τ^*} be the vector obtained by permuting two elements of $a_{\tau'}$, such that the $(k+1)$ -th element of a_{τ^*} coincides with the $(k+1)$ -th element of a_{τ} . Applying an argument analogous to the case $k = 2$, we obtain:

$$\sum_{i=1}^{k+1} a_{\tau^*[i]} \times w_{\sigma[i]} \leq \sum_{i=1}^{k+1} a_{\tau'[i]} \times w_{\sigma[i]}$$

Since at least the $(k+1)$ -th element of a_{τ^*} coincides with the $(k+1)$ -th element of a_{τ} , we obtain:

$$\sum_{i=1}^{k+1} a_{\tau[i]} \times w_{\sigma[i]} \leq \sum_{i=1}^{k+1} a_{\tau^*[i]} \times w_{\sigma[i]},$$

therefore, the condition also holds in this case.

Part 2. We show that the number of graphs in $\mathcal{G}_{[N]}$ such that $d_H(\mathcal{G}, \mathcal{G}^m) = h$, with $h \in (1, 2, \dots, N_e)$, is constant with respect to $\mathcal{G}^m$. A distance $d_H(\mathcal{G}, \mathcal{G}^m) = h$ implies that h entries of the adjacency matrix of $\mathcal{G}^m$ are modified. Since the graphs are labeled, the number of graphs such that $d_H(\mathcal{G}, \mathcal{G}^m) = h$ is given by the following quantity:

$$\binom{N_e}{h},$$

which is constant with respect to $\mathcal{G}^m$. The expected value $\mathbb{E}[d_H^2(\mathcal{G}, \mathcal{G}_k)]$ for a GCER model with parameters (W, α) is defined as follows:

$$\mathbb{E}[d_H^2(\mathcal{G}, \mathcal{G}_k)] = \sum_{i \in I} d_H^2(\mathcal{G}_i, \mathcal{G}_k) p_{\text{GCER}}(\mathcal{G}_i; W, \alpha)$$

where I is a set of indices representing the configurations $\mathcal{G}_i$ in $\mathcal{G}_{[N]}$, ordered such that $p_{\text{GCER}}(\mathcal{G}_i; W, \alpha) \geq p_{\text{GCER}}(\mathcal{G}_j; W, \alpha)$ for $i < j$. Since the number of graphs in $\mathcal{G}_{[N]}$ with distance $d_{\text{H}}(\mathcal{G}, \mathcal{G}^m) = h$ is constant with respect to $\mathcal{G}^m$, for every $h \in \{1, 2, \dots, N_e\}$, the vector of Hamming distances with respect to $\mathcal{G}_k$ can be obtained by permuting the entries of the vector of Hamming distances with respect to $\mathcal{G}^m$. This implies that

$$(d_{\text{H}}^2(\mathcal{G}_1, \mathcal{G}_k), d_{\text{H}}^2(\mathcal{G}_2, \mathcal{G}_k), \dots, d_{\text{H}}^2(\mathcal{G}_{|\mathcal{G}_{[N]}|}, \mathcal{G}_k))$$

is obtained by permuting the entries of the vector

$$(d_{\text{H}}^2(\mathcal{G}_1, \mathcal{G}^m), d_{\text{H}}^2(\mathcal{G}_2, \mathcal{G}^m), \dots, d_{\text{H}}^2(\mathcal{G}_{|\mathcal{G}_{[N]}|}, \mathcal{G}^m)).$$

Applying the result from Part (1) of the proof, it follows that the expected squared Hamming distance is minimized by the graph $\mathcal{G}^m$ and therefore $\mathcal{G}^m$ coincides with the Fréchet mean.

$$\mathbb{E}[d_{\text{H}}^2(\mathcal{G}, \mathcal{G}_k)] = \sum_{i \in I} d_{\text{H}}^2(\mathcal{G}_i, \mathcal{G}^m) p_{\text{GCER}}(\mathcal{G}_i; W, \alpha)$$

This result holds because the graphs closest to the location parameter W are also those closest to the mode of the distribution, since $\mathcal{G}^m$ is a discretization of the parameter W . Consequently, the graphs with smaller distance from $\mathcal{G}^m$ are assigned greater weight, as they have a higher probability of being observed. $\square$

Proposition 2.3. Let $\mathcal{G}_1$ and $\mathcal{G}_2$ be two graphs defined on the space $\mathcal{G}_{[N]}$. If the two graphs $\mathcal{G}_1$ and $\mathcal{G}_2$ are generated from the GCER model with $W \in \mathcal{W}_{[N]}$ and if $\alpha \in (0, 1/2)$, then $d_{\text{H}}(\mathcal{G}_1, W) > d_{\text{H}}(\mathcal{G}_2, W)$ implies that $p_{\text{GCER}}(\mathcal{G}_2; W, \alpha) > p_{\text{GCER}}(\mathcal{G}_1; W, \alpha)$, while $d_{\text{H}}(\mathcal{G}_1, W) = d_{\text{H}}(\mathcal{G}_2, W)$ implies that $p_{\text{GCER}}(\mathcal{G}_1; W, \alpha) = p_{\text{GCER}}(\mathcal{G}_2; W, \alpha)$.

Proof. Let $\mathcal{G}_1$ and $\mathcal{G}_2$ be in $\mathcal{G}_{[N]}$ and W in $\mathcal{W}_{[N]}$.

$$\begin{aligned}
\frac{p_{\text{GCER}}(\mathcal{G}_1; W, \alpha)}{p_{\text{GCER}}(\mathcal{G}_2; W, \alpha)} &= \frac{\left(\frac{1-\alpha}{\alpha}\right)^{-\sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}_1[i,j](1-2W[i,j])}}{\left(\frac{1-\alpha}{\alpha}\right)^{-\sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}_2[i,j](1-2W[i,j])}} \\
&= \left(\frac{1-\alpha}{\alpha}\right)^{\left(\sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}_2[i,j] - 2\sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}_2[i,j]W[i,j] + W[i,j]^2\right)} \\
&\quad \times \left(\frac{1-\alpha}{\alpha}\right)^{-\left(\sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}_1[i,j] - 2\sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}_1[i,j]W[i,j] + W[i,j]^2\right)} \\
&= \left(\frac{1-\alpha}{\alpha}\right)^{[d_{\text{H}}^2(\mathcal{G}_2, W) - d_{\text{H}}^2(\mathcal{G}_1, W)]}
\end{aligned}$$

Then, $\frac{p_{\text{GCER}}(\mathcal{G}_1; W, \alpha)}{p_{\text{GCER}}(\mathcal{G}_2; W, \alpha)} > 1$ if and only if $[d_{\text{H}}^2(\mathcal{G}_2, W) - d_{\text{H}}^2(\mathcal{G}_1, W)] \log\left(\frac{1-\alpha}{\alpha}\right) > 0$.

For $0 < \alpha < 1/2$, the quantity $\log\left(\frac{1-\alpha}{\alpha}\right)$ is positive, so that:

$$\frac{p_{\text{GCER}}(\mathcal{G}_1; W, \alpha)}{p_{\text{GCER}}(\mathcal{G}_2; W, \alpha)} > 1 \Leftrightarrow d_{\text{H}}^2(\mathcal{G}_2, W) > d_{\text{H}}^2(\mathcal{G}_1, W) \Leftrightarrow d_{\text{H}}(\mathcal{G}_2, W) > d_{\text{H}}(\mathcal{G}_1, W).$$

Following the same procedure, it is possible to prove the case $d_{\text{H}}(\mathcal{G}_2, W) = d_{\text{H}}(\mathcal{G}_1, W)$, which implies $p_{\text{GCER}}(\mathcal{G}_1; W, \alpha) = p_{\text{GCER}}(\mathcal{G}_2; W, \alpha)$. $\square$

2.2. Properties Related to the Scale Parameter

To better understand the effect of the scale parameter α on the probability distribution, it is useful to study its impact on the variability of the distribution. Since the distribution of each individual edge is known, its variance can be defined as follows

$$\begin{aligned}
\text{Var}(\mathcal{G}[i, j]) &= \left(1 + \left(\frac{\alpha}{1-\alpha}\right)^{2W[i,j]-1}\right)^{-1} \left(1 - \left(1 + \left(\frac{\alpha}{1-\alpha}\right)^{2W[i,j]-1}\right)^{-1}\right) \\
&= \frac{\left(\frac{\alpha}{1-\alpha}\right)^{2W[i,j]-1}}{\left(1 + \left(\frac{\alpha}{1-\alpha}\right)^{2W[i,j]-1}\right)^2}.
\end{aligned} \tag{2.2}$$

From equation (2.2), it can be observed that the variance of a single edge is minimized, namely $\text{Var}(\mathcal{G}[i, j]) = 0$, for $\alpha \rightarrow 0$ and $W[i, j] \in [0, 1] \setminus \{1/2\}$. As previously stated, when the scale parameter tends to zero, the resulting distribution is degenerate. Conversely, each

connection between nodes i and j reaches its maximum variability, equal to $\text{Var}(\mathcal{G}[i, j]) = \frac{1}{4}$, for $\alpha \rightarrow 1/2$, $W \in (0, 1)$, and when $\alpha \in (0, 1/2)$, $W = 1/2$, namely in the scenarios of maximum uncertainty for each edge.

Entropy is used as a metric to measure the overall variability of the GCER distribution. Entropy is a measure of variability that quantifies the uncertainty associated with the probability distribution and, in the context of the family $p_{\text{GCER}}(\cdot; W, \alpha)$, it can be interpreted as a measure of dispersion around the location parameter W . Entropy can be defined as follows:

$$H_{\mathcal{G}} = - \sum_{\mathcal{G} \in \mathcal{G}_{[N]}} p_{\text{GCER}}(\mathcal{G}; W, \alpha) \log(p_{\text{GCER}}(\mathcal{G}; W, \alpha)).$$

This expression can also be rewritten as:

$$H_{\mathcal{G}} = - \sum_{i=1}^{N-1} \sum_{j=i+1}^N H(\mathcal{G}[i, j]; W, \alpha) \quad (2.3)$$

where $H(\mathcal{G}[i, j]; W, \alpha)$ represents the entropy for each pair of nodes i and j .

Rewriting $H(\mathcal{G}[i, j]; W, \alpha)$ in expanded form yields the following expression:

$$\begin{aligned}
H(\mathcal{G}[i, j]) &= \left(1 - \left[1 + \left(\frac{\alpha}{1-\alpha}\right)^{(2W[i,j]-1)}\right]^{-1}\right) \log \left(1 - \left[1 + \left(\frac{\alpha}{1-\alpha}\right)^{(2W[i,j]-1)}\right]^{-1}\right) \\
&\quad + \left[1 + \left(\frac{\alpha}{1-\alpha}\right)^{(2W[i,j]-1)}\right]^{-1} \log \left(\left[1 + \left(\frac{\alpha}{1-\alpha}\right)^{(2W[i,j]-1)}\right]^{-1}\right) \\
&= \frac{\left(\frac{\alpha}{1-\alpha}\right)^{(2W[i,j]-1)}}{1 + \left(\frac{\alpha}{1-\alpha}\right)^{(2W[i,j]-1)}} \log \left(\frac{\alpha}{1-\alpha}\right)^{(2W[i,j]-1)} - \log \left(1 + \left(\frac{\alpha}{1-\alpha}\right)^{(2W[i,j]-1)}\right).
\end{aligned}$$

Substituting this quantity into the definition of entropy in (2.3) yields the overall entropy of the GCER distribution, namely:

$$\begin{aligned}
H_{\mathcal{G}} &= - \sum_{i=1}^{N-1} \sum_{j=i+1}^N \left[\frac{\left(\frac{\alpha}{1-\alpha}\right)^{2W[i,j]-1}}{1 + \left(\frac{\alpha}{1-\alpha}\right)^{2W[i,j]-1}} \log \left(\frac{\left(\frac{\alpha}{1-\alpha}\right)^{2W[i,j]-1}}{1 + \left(\frac{\alpha}{1-\alpha}\right)^{2W[i,j]-1}}\right) \right. \\
&\quad \left. + \frac{1}{1 + \left(\frac{\alpha}{1-\alpha}\right)^{2W[i,j]-1}} \log \left(\frac{1}{1 + \left(\frac{\alpha}{1-\alpha}\right)^{2W[i,j]-1}}\right) \right]. \tag{2.4}
\end{aligned}$$

To study how the scale parameter α affects the entropy of the distribution, the behavior of the entropy in the limiting cases of α is analyzed. From formula (2.4), it can be observed that for $\alpha \rightarrow 0$ and $W[i, j] \in [0, 1] \setminus \{1/2\}$

$$H_{\mathcal{G}} = 0.$$

In other words, as α tends to zero, the graphs exhibit maximum concentration around the location parameter W .

Conversely, for $\alpha \rightarrow 1/2$, $W[i, j] \in (0, 1)$ and for $W[i, j] = 1/2$, $\alpha \in (0, 1/2)$

$$H_{\mathcal{G}} = \frac{N(N-1)}{2}.$$

In this scenario, the dispersion around the weighted graph W is maximal, since for each edge $H[\mathcal{G}[i, j]] = 1/2$.

Proposition 2.4. For any fixed location parameter $W \in \mathcal{W}_{[N]}$, the entropy associated with

the family $p_{\text{GCER}}(\cdot; W, \alpha)$ is a monotonic function of α .

Proof. To prove Proposition 4.2.1, we compute the derivative of equation (2.4) with respect to the scale parameter α

$$\frac{dH_{\mathcal{G}}}{d\alpha} = -\log\left(\frac{\alpha}{1-\alpha}\right) \sum_{i=1}^{N-1} \sum_{j=i+1}^N \frac{(2W[i,j] - 1)^2}{\left(1 + \left(\frac{\alpha}{1-\alpha}\right)^{(2W[i,j]-1)}\right)^2} \left(\frac{\alpha}{1-\alpha}\right)^{(2W[i,j]-1)-1}.$$

Since $\alpha \in (0, 1/2)$, the derivative of the entropy is positive, which implies that the entropy increases as the scale parameter α increases. □

3. Bayesian Inference for the GCER Model

Having defined the theoretical aspects and the properties characterizing the GCER model, the focus now shifts to the estimation of the parameters α and W through Bayesian inference techniques.

Let $\{\mathcal{G}_1, \dots, \mathcal{G}_n\}$ be a set of observations defined on $\mathcal{G}_{[N]}$ and let W be defined on $\mathcal{W}_{[N]}$. The posterior probability distribution to be estimated is the following:

$$p(W, \alpha \mid \mathcal{G}_1, \dots, \mathcal{G}_n) \propto p(W \mid \alpha) p(\alpha) \prod_{i=1}^n p(\mathcal{G}_i \mid W, \alpha), \quad (3.1)$$

where $p(\mathcal{G}_i \mid W, \alpha)$ corresponds to the likelihood function of the GCER model.

For simplicity, the scale parameter α is assumed to be known, so that the inferential analysis focuses only on the location parameter W . In the absence of specific prior knowledge about the structure of W , a uniform prior distribution defined on the interval $[0, 1]$ is assigned to each of its components. This choice simplifies the computation of the posterior distribution, which can be expressed as:

$$p(W \mid \mathcal{G}_1, \dots, \mathcal{G}_n) \propto \prod_{i=1}^n p(\mathcal{G}_i \mid W, \alpha). \quad (3.2)$$

Rewriting equation (3.2) in expanded form yields:

$$p(W \mid \mathcal{G}_1, \dots, \mathcal{G}_n) \propto \left(\frac{\alpha}{1 - \alpha} \right)^{\sum_{i=1}^{N-1} \sum_{j=i+1}^N \sum_{k=1}^n \mathcal{G}_k[i,j] (1 - 2W[i,j])} \prod_{i=1}^{N-1} \prod_{j=i+1}^N \left(1 + \left(\frac{\alpha}{1 - \alpha} \right)^{1 - 2W[i,j]} \right)^{-n}. \quad (3.3)$$

From equation (3.3), it can be observed that the posterior distribution does not belong to a known family of distributions and has a rather complex analytical form. To sample the location parameter W from its posterior distribution, it is therefore necessary to resort to simulation-based algorithmic methods, such as the Metropolis–Hastings Random Walk, a Markov chain Monte Carlo algorithm used to sample from complex posterior distributions. The basic idea of the method consists of proposing, at each iteration, a new candidate $W'[i, j]$ obtained by sampling from a normal distribution centered at the value of $W[i, j]$ from the previous iteration and with fixed variance equal to σ^2 . The new candidate is accepted or rejected based on the value of the acceptance ratio defined as:

$$r = \min\left\{1, \frac{p(W[i, j] | W'[i, j])}{p(W'[i, j] | W[i, j])}\right\}$$

where $W[i, j]$ represents the current value and $W'[i, j]$ the new candidate.

The choice of the parameter σ^2 of the Gaussian kernel plays a crucial role, as it directly affects the acceptance probability of the proposed values of $W[i, j]$ in the Metropolis–Hastings Random Walk method. If excessively large values of σ^2 are considered, the acceptance rate may be too low because the algorithm, by taking excessively large steps, is unable to efficiently explore the entire parameter space; conversely, if excessively small values of σ^2 are considered, the acceptance rate may be very high because the algorithm, by taking very small steps, tends to repeatedly accept similar values.

These two scenarios are problematic for ensuring the convergence of the algorithm; for this reason, it is necessary to set the variability parameter efficiently.

4. Bayesian Inference for the CER Model with a GCER Prior

This section aims to introduce different hierarchical Bayesian models in order to conduct an inferential analysis of the location parameter $\mathcal{G}^m$ and the scale parameter α characterizing the CER model. This analysis is conducted under the assumption that the sampling distribution and the prior distribution specified for the Fréchet mean have different functional forms. Regarding the likelihood function, the sample is assumed to follow a CER distribution, while a GCER distribution is assigned as the prior distribution for the location parameter $\mathcal{G}^m$.

Let $N \in \mathbb{N}$ and let d_H be the Hamming distance defined on $\mathcal{G}_{[N]}$. Let $\{\mathcal{G}_1, \dots, \mathcal{G}_n\}$ be a set of observations defined on $\mathcal{G}_{[N]}$. The distribution of interest to be estimated is the following:

$$p(\mathcal{G}^m, \alpha \mid \mathcal{G}_1, \dots, \mathcal{G}_n) = p(\mathcal{G}^m \mid \alpha) p(\alpha) \prod_{i=1}^n p(\mathcal{G}_i \mid \mathcal{G}^m, \alpha) \quad (4.1)$$

where:

$$p(\mathcal{G}_i \mid \mathcal{G}^m, \alpha) = \alpha^{d_H(\mathcal{G}_i, \mathcal{G}^m)} (1 - \alpha)^{\frac{(N-1)N}{2} - d_H(\mathcal{G}_i, \mathcal{G}^m)}. \quad (4.2)$$

Expressions (4.1) and (4.2) represent the key elements for defining different specifications of hierarchical Bayesian models.

4.1. GCER Distribution as a Conjugate Prior in a CER Model

Proposition 4.1. The GCER distribution is a conjugate prior for the location parameter $\mathcal{G}^m$ of the CER model with known scale parameter α .

Before proceeding with the proof, it is necessary to clarify some aspects of the structure of the hierarchical model under consideration. Specifically, the reference model is the following:

$$\mathcal{G}_i \mid \mathcal{G}^m, \alpha \stackrel{\text{iid}}{\sim} \text{CER}(\mathcal{G}^m, \alpha)$$

$$\mathcal{G}^m \sim \text{GCER}(W_0, \tau_{\nu_0}(\alpha)),$$

where $\nu_0 > 0$ and $W_0 \in \mathcal{W}_{[N]}$ and the function $\tau_{\nu_0} : [0, 1/2] \rightarrow [0, 1/2]$ is defined as:

$$\tau_{\nu_0}(\alpha) = \frac{\alpha^{\nu_0}}{(1 - \alpha)^{\nu_0} + \alpha^{\nu_0}}.$$

Through τ_{ν_0} , the parameter α is transformed to define the scale parameter of the GCER prior distribution for $\mathcal{G}^m$, so that it does not necessarily coincide with that of the CER kernel, which occurs when $\nu_0 = 1$.

The quantity ν_0 can be interpreted as the size of the ideal sample on which the prior belief about $\mathcal{G}^m$ is based. It can be observed that for large values of ν_0 , the scale parameter becomes increasingly smaller and, consequently, the realizations become highly concentrated around the prior base measure W_0 .

Proof. Under the assumption that the scale parameter α is known, the posterior distribution of $\mathcal{G}^m$ is proportional to the product of the likelihood function and the prior distribution for $\mathcal{G}^m$. Formally:

$$p(\mathcal{G}^m \mid \mathcal{G}_1, \dots, \mathcal{G}_n) \propto p(\mathcal{G}_1, \dots, \mathcal{G}_n \mid \mathcal{G}^m) p(\mathcal{G}^m).$$

The likelihood function is given by equation (4.2).

Regarding the prior distribution, it takes the following form:

$$p(\mathcal{G}^m) \propto \left(\frac{\tau_{\nu_0}(\alpha)}{1 - \tau_{\nu_0}(\alpha)} \right)^{\sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}^m[i,j](1-2W_0[i,j])}$$

where, by expanding the quantity $\tau_{\nu_0}(\alpha)$, it can be rewritten as:

$$p(\mathcal{G}^m) \propto \left(\frac{\alpha}{1-\alpha} \right)^{\nu_0 \sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}^m[i,j](1-2W_0[i,j])}.$$

Having defined the quantities required to compute the posterior distribution of $\mathcal{G}^m$, the analytical steps for deriving it are presented below.

$$\begin{aligned} p(\mathcal{G}^m \mid \mathcal{G}_1, \dots, \mathcal{G}_n) &\propto p(\mathcal{G}_1, \dots, \mathcal{G}_n \mid \mathcal{G}^m) p(\mathcal{G}^m) \\ &= \alpha^{\nu_0 \sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}^m[i,j](1-2W_0[i,j]) + \sum_{k=1}^n d_H(\mathcal{G}^m, \mathcal{G}_k)} \\ &\quad \times (1-\alpha)^{\frac{nN(N-1)}{2} - \sum_{k=1}^n d_H(\mathcal{G}^m, \mathcal{G}_k) - \nu_0 \sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}^m[i,j](1-2W_0[i,j])} \\ &\propto \left(\frac{\alpha}{1-\alpha} \right)^{\sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}^m[i,j](\nu_0 - 2\nu_0 W_0[i,j] + n - 2 \sum_{k=1}^n \mathcal{G}_k[i,j])} \\ &= \left(\frac{\alpha}{1-\alpha} \right)^{\sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}^m[i,j](\nu_0 + n) \left(1 - 2 \frac{\nu_0 W_0[i,j] + \sum_{k=1}^n \mathcal{G}_k[i,j]}{\nu_0 + n} \right)} \\ &= \left(\frac{\alpha}{1-\alpha} \right)^{\nu' \sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}^m[i,j](1-2W'[i,j])}. \end{aligned}$$

From the expression of the posterior distribution of $\mathcal{G}^m$, it can be observed that the GCER distribution is a conjugate prior for the CER model. Therefore, it follows that:

$$\mathcal{G}^m \mid \mathcal{G}_1, \dots, \mathcal{G}_n \sim \text{GCER}(W', \tau_{\nu'}(\alpha)) \quad (4.3)$$

where

$$\tau_{\nu'}(\alpha) = \tau_{\nu_0 + n}(\alpha)$$

and

$$W' = \frac{\nu_0}{\nu_0 + n} W_0 + \frac{n}{\nu_0 + n} \frac{\sum_{k=1}^n \mathcal{G}_k}{n} = \frac{\nu_0}{\nu_0 + n} W_0 + \frac{n}{\nu_0 + n} \bar{\mathcal{G}}.$$

□

From the structure of the posterior distribution of $\mathcal{G}^m$, it can be observed that the new scale parameter $\tau_{\nu'}(\alpha)$ depends on both the sample size n and ν_0 .

Regarding the location parameter, it is defined as a linear combination of the matrix W_0 , which represents the prior weights associated with each pair of nodes i and j , and the matrix

$\bar{\mathcal{G}}$, whose elements $\bar{\mathcal{G}}[i, j]$ correspond to the sample mean of the connections for each pair of nodes. The weights of this linear combination are proportional to the parameter ν_0 and the sample size n , respectively.

Thanks to the conjugacy property of the GCER distribution, it is possible to define the posterior probability for $\mathcal{G}^m$ of observing a connection between the pair of nodes i and j . This probability is defined as follows:

$$\Pr(\mathcal{G}^m[i, j] = 1 \mid \mathcal{G}_1, \dots, \mathcal{G}_n) = \left[1 + \left(\frac{\alpha}{1 - \alpha} \right)^{\nu'(2W'[i, j] - 1)} \right]^{-1}.$$

Consequently, conditionally on the observations $\mathcal{G}_1, \dots, \mathcal{G}_n$, the variable $\mathcal{G}^m[i, j]$ follows a Bernoulli distribution of the following form:

$$\mathcal{G}^m[i, j] \mid \mathcal{G}_1, \dots, \mathcal{G}_n \sim \text{Bernoulli} \left(\left[1 + \left(\frac{\alpha}{1 - \alpha} \right)^{\nu'(2W'[i, j] - 1)} \right]^{-1} \right).$$

4.2. Unknown Scale Parameter α

Assuming that the parameter α is unknown, two distinct specifications can be defined for the prior distributions of the parameters $(\mathcal{G}^m, \alpha)$ of a CER model.

4.2.1. Prior with Dependent Components

The reference model is the following:

$$\begin{aligned} \mathcal{G}_i \mid \mathcal{G}^m &\stackrel{\text{iid}}{\sim} \text{CER}(\mathcal{G}^m, \alpha), \\ \mathcal{G}^m \mid \alpha &\sim \text{GCER}(W_0, \tau_{\nu_0}(\alpha)), \\ \alpha &\sim \text{TBeta}(1/2, a_0, b_0). \end{aligned}$$

In this scenario, a truncated Beta distribution defined on the interval $[0, 1/2]$ is chosen as the prior distribution for the scale parameter α , since it is convenient from an analytical point of view and ensures that the support of the parameter α is restricted to the interval between 0

and $1/2$.

The joint posterior distribution of the parameters $(\mathcal{G}^m, \alpha)$ is proportional to the product of the likelihood function, the prior distribution assigned to the location parameter $\mathcal{G}^m$, and the prior distribution assigned to the scale parameter α . Formally:

$$p(\mathcal{G}^m, \alpha \mid \mathcal{G}_1, \dots, \mathcal{G}_n) \propto p(\mathcal{G}_1, \dots, \mathcal{G}_n \mid \mathcal{G}^m, \alpha) p(\mathcal{G}^m \mid \alpha) p(\alpha).$$

The likelihood function is given by equation (4.2), while the prior distribution for $\mathcal{G}^m$ is:

$$p(\mathcal{G}^m \mid \alpha) = \left(\frac{\alpha}{1-\alpha} \right)^{\nu_0 \sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}^m[i,j] (1-2W_0[i,j])} \frac{1}{\prod_{i=1}^{N-1} \prod_{j=i+1}^N \left(1 + \left(\frac{\alpha}{1-\alpha} \right)^{\nu_0 (1-2W_0[i,j])} \right)}.$$

Regarding the scale parameter α , the following truncated Beta prior distribution is assigned:

$$p(\alpha) = \begin{cases} \frac{\alpha^{a_0-1} (1-\alpha)^{b_0-1}}{\int_0^{1/2} \alpha^{a_0-1} (1-\alpha)^{b_0-1} d\alpha}, & \text{if } \alpha \in [0, 1/2], \\ 0 & \text{otherwise.} \end{cases}$$

Having defined the quantities required to compute the joint posterior distribution of the parameters $(\mathcal{G}^m, \alpha)$, the analytical steps for deriving it are presented below:

$$\begin{aligned} p(\mathcal{G}^m, \alpha \mid \mathcal{G}_1, \dots, \mathcal{G}_n) &\propto p(\mathcal{G}_1, \dots, \mathcal{G}_n \mid \mathcal{G}^m, \alpha) p(\mathcal{G}^m \mid \alpha) p(\alpha) \\ &= \alpha^{\sum_{k=1}^n d_H(\mathcal{G}^m, \mathcal{G}_k) + \nu_0 \sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}^m[i,j] (1-2W_0[i,j]) + a_0 - 1} \\ &\quad \times (1-\alpha)^{\frac{nN(N-1)}{2} - \sum_{k=1}^n d_H(\mathcal{G}^m, \mathcal{G}_k) - \nu_0 \sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}^m[i,j] (1-2W_0[i,j]) + b_0 - 1} \\ &\quad \times \frac{1}{\prod_{i=1}^{N-1} \prod_{j=i+1}^N \left(1 + \left(\frac{\alpha}{1-\alpha} \right)^{\nu_0 (1-2W_0[i,j])} \right)} \end{aligned}$$

$$\begin{aligned}
&= \left(\frac{\alpha}{1-\alpha} \right)^{\sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}^m[i,j](\nu_0+n)} \left(1 - 2 \frac{\nu_0 W_0[i,j] + \sum_{k=1}^n \mathcal{G}_k[i,j]}{\nu_0+n} \right) \\
&\quad \times \alpha^{\sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}_k[i,j] + a_0 - 1} \\
&\quad \times (1-\alpha)^{\frac{nN(N-1)}{2} - \sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}_k[i,j] + b_0 - 1} \\
&\quad \times \frac{1}{\prod_{i=1}^{N-1} \prod_{j=i+1}^N \left(1 + \left(\frac{\alpha}{1-\alpha} \right)^{\nu_0(1-2W[i,j])} \right)}.
\end{aligned}$$

From the expression of the joint posterior distribution, it can be observed that, when α is unknown, the conjugacy property does not hold. For this reason, in order to sample from the posterior distribution of $(\mathcal{G}^m, \alpha)$, it is necessary to resort to simulation-based algorithmic methods, such as a Gibbs sampler combined with a Metropolis–Hastings Random Walk. The basic idea of the Gibbs sampler algorithm is to sample from the joint posterior distribution using the full conditional distributions of the parameters involved.

From the expression of the posterior distribution, it is possible to derive the full conditional distributions for the parameters $\mathcal{G}^m$ and α . The full conditional distribution for $\mathcal{G}^m$ corresponds to the GCER model with updated parameters, while the full conditional distribution of the scale parameter α has a more complex structure from which it is difficult to sample directly. Its functional form is the following:

$$\begin{aligned}
p(\alpha \mid \mathcal{G}_1, \dots, \mathcal{G}_n, \mathcal{G}^m) &\propto \alpha^{\sum_{k=1}^n d_H(\mathcal{G}^m, \mathcal{G}_k) + \nu_0 \sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}^m[i,j](1-2W_0[i,j]) + a_0 - 1} \\
&\quad \times (1-\alpha)^{\frac{nN(N-1)}{2} - \sum_{k=1}^n d_H(\mathcal{G}^m, \mathcal{G}_k) - \nu_0 \sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}^m[i,j](1-2W_0[i,j]) + b_0 - 1} \\
&\quad \times \frac{1}{\prod_{i=1}^{N-1} \prod_{j=i+1}^N \left(1 + \left(\frac{\alpha}{1-\alpha} \right)^{\nu_0(1-2W_0[i,j])} \right)}.
\end{aligned}$$

In the Gibbs sampling algorithm, to update the adjacency matrix associated with $\mathcal{G}^m$ at each iteration, it is sufficient to sample from the GCER distribution

$$p(\mathcal{G}^m \mid \mathcal{G}_1, \dots, \mathcal{G}_n, \alpha) \propto \left(\frac{\alpha}{1-\alpha} \right)^{\sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}^m \left(1 - 2 \frac{\nu_0 W_0[i,j] + \sum_{k=1}^n \mathcal{G}_k[i,j]}{\nu_0+n} \right)},$$

with updated parameters $\tau_{\nu'}(\alpha) = \tau_{\nu_0+n}(\alpha)$ and $W' = \frac{\nu_0}{\nu_0+n} W_0 + \frac{n}{\nu_0+n} \bar{\mathcal{G}}$.

To update the scale parameter α , instead, the Metropolis–Hastings Random Walk method is used. At each iteration, a new value of α is proposed, generated from a normal distribution centered at the current value, with a given variance σ^2 . The newly proposed parameter is accepted or rejected based on the value of the acceptance ratio, defined as follows:

$$r = \min\left(1, \frac{p(\alpha^{\text{proposed}} \mid \mathcal{G}_1, \dots, \mathcal{G}_n)}{p(\alpha^{\text{current}} \mid \mathcal{G}_1, \dots, \mathcal{G}_n)}\right).$$

The choice of the parameter σ^2 of the Gaussian kernel plays a crucial role, as it directly affects the acceptance probability of the proposed values of α in the Metropolis–Hastings Random Walk method. To ensure convergence of the algorithm, the parameter σ^2 must be chosen so that the acceptance rate falls within an optimal range. Simulation studies suggest an ideal acceptance rate between 20% and 30%. See also Roberts & Rosenthal (2001).

4.2.2. Prior with Independent Components

In this context, it is assumed that the scale parameter characterizing the likelihood function differs from that of the prior distribution for $\mathcal{G}^m$.

Consequently, the model can be specified as follows:

$$\begin{aligned}\mathcal{G}_i \mid \mathcal{G}^m &\stackrel{\text{iid}}{\sim} \text{CER}(\mathcal{G}^m, \alpha), \\ \mathcal{G}^m &\sim \text{GCER}(W_0, \alpha_0), \\ \alpha &\sim \text{TBeta}(1/2, a_0, b_0).\end{aligned}$$

The joint posterior distribution of the parameters $(\mathcal{G}^m, \alpha)$ to be estimated is defined as follows:

$$p(\mathcal{G}^m, \alpha \mid \mathcal{G}_1, \dots, \mathcal{G}_n) \propto p(\mathcal{G}_1, \dots, \mathcal{G}_n \mid \mathcal{G}^m, \alpha) p(\mathcal{G}^m) p(\alpha),$$

which, when rewritten in expanded form, has the following functional form:

$$\begin{aligned}
p(\mathcal{G}^m, \alpha \mid \mathcal{G}_1, \dots, \mathcal{G}_n) &\propto \alpha^{\sum_{k=1}^n d_H(\mathcal{G}^m, \mathcal{G}_k) + a_0 - 1} \\
&\times (1 - \alpha)^{\frac{nN(N-1)}{2} - \sum_{k=1}^n d_H(\mathcal{G}^m, \mathcal{G}_k) + b_0 - 1} \\
&\times \left(\frac{\alpha_0}{1 - \alpha_0} \right)^{\sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}^m[i,j](1-2W_0[i,j])} \\
&\times \frac{1}{\prod_{i=1}^{N-1} \prod_{j=i+1}^N \left(1 + \left(\frac{\alpha_0}{1-\alpha_0} \right)^{(1-2W_0[i,j])} \right)}.
\end{aligned}$$

Also in this case, the prior distribution is not conjugate to the likelihood function. For this reason, in order to sample from the posterior distribution of $(\mathcal{G}^m, \alpha)$, it is necessary to resort to simulation-based algorithmic methods, such as the Gibbs sampler.

From the expression of the posterior distribution, it is possible to derive the full conditional distributions for the parameters $\mathcal{G}^m$ and α .

The full conditional distribution for $\mathcal{G}^m$ has the following functional form:

$$\begin{aligned}
p(\mathcal{G}^m \mid \mathcal{G}_1, \dots, \mathcal{G}_n, \alpha) &\propto \left(\frac{\alpha}{1 - \alpha} \right)^{n \sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}^m[i,j] \left(1 - 2 \sum_{k=1}^n \frac{\mathcal{G}_k[i,j]}{n} \right)} \\
&\times \left(\frac{\alpha_0}{1 - \alpha_0} \right)^{\sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathcal{G}^m[i,j](1-2W_0[i,j])}.
\end{aligned}$$

The full conditional distribution for the scale parameter α , instead, has a known form and corresponds to a truncated Beta distribution with updated parameters:

$$\alpha \mid \mathcal{G}_1, \dots, \mathcal{G}_n, \mathcal{G}^m \sim \text{TBeta} \left(1/2, \sum_{k=1}^n d_H(\mathcal{G}^m, \mathcal{G}_k) + a_0, \frac{nN(N-1)}{2} - \sum_{k=1}^n d_H(\mathcal{G}^m, \mathcal{G}_k) + b_0 \right).$$

5. Discriminant Analysis as an Application Example

The multiple-network setting gives rise to several potential applications, one of which concerns the definition of decision rules for classifying new graphs, whose class membership is unknown, based on their structural properties.

At this stage of the analysis, the focus is on the classification of observations distributed according to the CER model, considering the cases in which the observations are characterized by homoscedasticity and heteroscedasticity.

For this purpose, the Bayesian classifier defines the decision rule underlying discriminant analysis, according to which a new subject is assigned to the category corresponding to the highest posterior probability. Formally, assuming that the observations are divided into K categories:

$$\Pr(Y = k \mid \mathcal{G}) = \max_{k \in \{1, \dots, K\}} \Pr(Y = k \mid \mathcal{G}),$$

where the term $\Pr(Y = k \mid \mathcal{G})$ represents the posterior probability that the graph $\mathcal{G}$ belongs to class k . This quantity is computed using Bayes' theorem and can be written as:

$$\Pr(Y = k \mid \mathcal{G}) = \frac{\Pr(Y = k)p(\mathcal{G} \mid Y = k)}{\sum_{k=1}^K \Pr(Y = k)p(\mathcal{G} \mid Y = k)}.$$

The Bayes classifier depends on the prior distribution $\Pr(Y = k)$, namely the prior probability of belonging to the k -th class, and on the likelihood function $p(\mathcal{G} \mid Y = k)$, which describes how the data are distributed within each class.

The latter quantity corresponds to the CER model, so that the likelihood function can be rewritten in expanded form as:

$$p(\mathcal{G} \mid Y = k) = \alpha^{d_H(\mathcal{G}_k^m, \mathcal{G})} (1 - \alpha)^{\frac{N(N-1)}{2} - d_H(\mathcal{G}_k^m, \mathcal{G})}.$$

5.1. Homoscedasticity Assumption

In this scenario, the CER model is assumed to be parameterized by a location parameter $\mathcal{G}^m$ specific to each k -th class and a scale parameter α common to all classes. When the degree of dispersion of the observations around the location parameter does not depend on class membership, the homoscedasticity condition holds. In notation, for an observation $\mathcal{G}$ assigned to the k -th class:

$$\mathcal{G} \sim \text{CER}(\mathcal{G}_k^m, \alpha).$$

Initially, assume that the observations are divided into two classes ($K = 2$). A new subject will belong to the first group if the following condition holds:

$$\Pr(Y = 1)p(\mathcal{G} \mid Y = 1) > \Pr(Y = 2)p(\mathcal{G} \mid Y = 2),$$

which, under the assumption of uniform prior distributions, can be rewritten as follows:

$$p(\mathcal{G} \mid Y = 1) > p(\mathcal{G} \mid Y = 2).$$

Substituting the appropriate quantities into the inequality and applying a logarithmic transformation for analytical convenience yields:

$$d_H(\mathcal{G}_1^m, \mathcal{G}) \log(\alpha) - d_H(\mathcal{G}_1^m, \mathcal{G}) \log(1 - \alpha) > d_H(\mathcal{G}_2^m, \mathcal{G}) \log(\alpha) - d_H(\mathcal{G}_2^m, \mathcal{G}) \log(1 - \alpha).$$

The probability that a graph belongs to class $k = 1$ is greater than the probability that it belongs to class $k = 2$ if the Hamming distance between the graph and $\mathcal{G}_1^m$ is smaller than

that between the graph and $\mathcal{G}_2^m$, that is:

$$d_H(\mathcal{G}_1^m, \mathcal{G}) < d_H(\mathcal{G}_2^m, \mathcal{G}).$$

Extending this result to the case in which K classes are considered, the decision rule can be defined as follows:

$$Class(\mathcal{G}) = \underset{k}{\operatorname{argmin}} d_H(\mathcal{G}, \mathcal{G}_k^m). \quad (5.1)$$

In conclusion, a new graph is assigned to the class whose concentration parameter minimizes the Hamming distance from the graph itself.

5.2. Heteroscedasticity Assumption

Consider the scenario in which the graphs are distributed according to a CER model with class-specific location and scale parameters, that is:

$$\mathcal{G} \sim \text{CER}(\mathcal{G}_k^m, \alpha_k).$$

In this case, the observations are heteroscedastic, meaning that their degree of dispersion around the parameter $\mathcal{G}^m$ is no longer common to all classes but is defined specifically for each of them.

Initially, consider two classes $K = 2$. A new graph will belong to the first class if the following condition holds:

$$\Pr(Y = 1)p(\mathcal{G} \mid Y = 1) > \Pr(Y = 2)p(\mathcal{G} \mid Y = 2).$$

Under the assumption of uniform prior distributions and rewriting the condition explicitly, we obtain:

$$\alpha_1^{d_H(\mathcal{G}_1^m, \mathcal{G})} (1 - \alpha_1)^{\frac{N(N-1)}{2} - d_H(\mathcal{G}_1^m, \mathcal{G})} > \alpha_2^{d_H(\mathcal{G}_2^m, \mathcal{G})} (1 - \alpha_2)^{\frac{N(N-1)}{2} - d_H(\mathcal{G}_2^m, \mathcal{G})}.$$

Generalizing this result to the case in which a larger number of classes, equal to K , is considered, the decision rule is defined as follows:

$$Class(\mathcal{G}) = \underset{k}{\operatorname{argmax}} \alpha_k^{d_H(\mathcal{G}_k^m, \mathcal{G})} (1 - \alpha_k)^{\frac{N(N-1)}{2} - d_H(\mathcal{G}_k^m, \mathcal{G})}, \quad (5.2)$$

which states that a new graph will be assigned to the class corresponding to the maximum likelihood.

6. Simulation Study

Having defined the theoretical aspects and the main techniques used to estimate the parameters of the CER and GCER models, the analysis proceeds with the presentation of a simulation study aimed at comparing three different model configurations in order to evaluate the flexibility and adaptability of the GCER model compared with the CER model.

The hierarchical Bayesian models considered are the following:

- **CER model with a CER prior on $\mathcal{G}^m$:**

$$\mathcal{G}_i \mid \mathcal{G}^m, \alpha \sim \text{CER}(\mathcal{G}^m, \alpha), \quad \mathcal{G}^m \sim \text{CER}(\mathcal{G}_0, \alpha).$$

In this first scenario, the same functional form is used to model the observations and the scale parameter $\mathcal{G}^m$.

- **CER model with a GCER prior on $\mathcal{G}^m$:**

$$\mathcal{G}_i \mid \mathcal{G}^m, \alpha \sim \text{CER}(\mathcal{G}^m, \alpha), \quad \mathcal{G}^m \sim \text{GCER}(W_0, \alpha).$$

In this second scenario, different functional forms are used to model the observations and the scale parameter $\mathcal{G}^m$.

- **GCER model with a uniform prior on W :**

$$\mathcal{G}_i \mid W, \alpha \sim \text{GCER}(W, \alpha), \quad W \sim \mathcal{U}(0, 1).$$

In this last case, the observations are modeled using the generalized GCER model, assigning a non-informative prior to the location parameter W .

In the three specifications introduced above, the scale parameter α is assumed to be known. To estimate this parameter efficiently, an ‘‘Empirical Bayes’’ approach was used. The basic

idea consists of selecting the parameter α that maximizes the joint distribution of the data, or marginal likelihood, formally:

$$\hat{\alpha} = \operatorname{argmax}_{\alpha \in (0, 1/2)} p(\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_n).$$

The joint distribution of the data $p(\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_n)$ varies according to the model under consideration:

- **CER model with a CER prior or CER model with a GCER prior**

In the case of a CER model, the joint distribution is obtained by marginalizing over $\mathcal{G}^m$

$$p(\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_n) = \sum_{\mathcal{G}^m \in \mathcal{G}_{[N]}} p(\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_n \mid \mathcal{G}^m) p(\mathcal{G}^m). \quad (6.1)$$

- **GCER model with a uniform prior on W**

In the case of a GCER model, the joint distribution is obtained by marginalizing over the weighted graph W

$$p(\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_n) = \int_{[0,1]^{N(N-1)/2}} p(\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_n \mid W) p(W) dW. \quad (6.2)$$

In the case of a CER model, analytically deriving the joint distribution from equation (6.1) is very complex, since marginalizing over $\mathcal{G}^m$ requires summing over all possible configurations of the location parameter, whose number grows exponentially as N increases and is equal to $2^{\frac{N(N-1)}{2}}$. Similarly, in the case of the GCER model, solving expression (6.2) is very complex because of the high dimensionality of the integration space. For this reason, in both cases, the joint distribution is estimated using Monte Carlo simulation-based methods.

The Monte Carlo approach consists of approximating equations (6.1) and (6.2) by sampling the parameters $\mathcal{G}^m$ and W s times, with s sufficiently large, from their respective prior distributions. In the case of the CER model, the joint distribution is approximated as follows:

$$p(\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_n) \approx \frac{1}{s} \sum_{i=1}^s p(\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_n \mid \mathcal{G}_i^m)$$

where $\mathcal{G}_1^m, \dots, \mathcal{G}_s^m$ are the samples generated from the prior distribution $p(\mathcal{G}^m)$. Similarly, for the GCER model:

$$p(\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_n) \approx \frac{1}{s} \sum_{i=1}^s p(\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_n \mid W_i)$$

where $\mathcal{G}_1^m, \dots, \mathcal{G}_s^m$ are the samples generated from the prior distribution $p(W)$. In both cases, the resulting Monte Carlo approximation is a quantity that depends on the scale parameter α . Numerical optimization methods are used to identify the value of the parameter α that maximizes the distribution $p(\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_n)$.

Focusing now on the estimation of the location parameters $\mathcal{G}^m$ and W , the following approaches were used in the three configurations:

- **CER model with a CER prior**

The location parameter $\mathcal{G}^m$ was estimated by implementing a Gibbs sampling algorithm with 10000 iterations, of which 1000 were used as burn-in.

- **CER model with a GCER prior**

In the case of a CER model with a GCER prior, the parameter $\mathcal{G}^m$ was estimated by exploiting the conjugacy property of the GCER distribution with respect to the CER model.

- **GCER model with a uniform prior on W**

In the case of a GCER model, the parameter W was estimated by implementing a Metropolis–Hastings Random Walk with 1000 iterations, of which 100 were used as burn-in.

Having defined the estimation strategies used to estimate the parameters of the three model configurations, two datasets with $n = 1000$ observations and $N = 10$ nodes are simulated. The two datasets were simulated considering two different scenarios: in the first case, the observations were generated from a CER model and subsequently from a GCER model.

The aim of the analysis is to evaluate the goodness-of-fit performance of the three models under the two simulation scenarios. From the comparison of the three model configurations applied to the two datasets, the CER model is expected to provide the best fit to data generated from the same functional form, while maintaining performance similar to that of the

GCER model with a uniform prior. Conversely, for data generated from the GCER model, a substantial difference is expected between the performance of the GCER model and that of the CER model.

To evaluate the goodness of fit of the three models, the Log Pseudo Marginal Likelihood was chosen as the evaluation metric (Geisser & Eddy, 1979). The Log Pseudo Marginal Likelihood is a Bayesian criterion used for model selection and is defined as follows:

$$\text{LPML} = \sum_{i=1}^n \log(\text{CPO}_i),$$

where CPO_i corresponds to $f_i(y_i \mid y_{-i}, M)$, namely the predictive density estimated based on all observations, given model M , excluding the i -th observation. To compute the LPML, it can be shown that, by sampling from the posterior distribution of the parameter vector θ , the quantity CPO_i can be defined as:

$$(\text{CPO}_i)^{-1} = \frac{1}{s} \sum_{k=1}^s \frac{1}{f_i(y_i \mid \theta^k, M)},$$

where:

- $\theta = (\theta_1, \dots, \theta_m)$ represents the model parameter vector with m components
- θ^k is the value of the parameter vector at step k of the sampling chain
- $f_i(y_i \mid \theta^k, M)$ represents the likelihood function for the i -th observation, given model M
- s corresponds to the number of generated posterior samples.

Higher LPML values indicate that the model provides a better fit to the data; therefore, the model with the highest Log Pseudo Marginal Likelihood will be selected.

The goodness-of-fit performance of the three model configurations applied to the two simulation scenarios is reported in Table 6.1.

Dataset Origin	Model	LPML
Simulated from CER model	CER with CER prior	-22562.06
	CER with GCER prior	-22562.06
	GCER with uniform prior	-22581.36
Simulated from GCER model	CER with CER prior	-30018.25
	CER with GCER prior	-30010.25
	GCER with uniform prior	-29551.02

Table 6.1: LPML values for the three model configurations according to the dataset origin.

The results reported in the table are consistent with expectations. It can be observed that, in the first scenario, in which the data are generated from the CER model, the best performance in terms of LPML is obtained by adopting the same model used to generate the data. The performance of the GCER model, however, is close to that of the CER model, highlighting its flexibility.

Regarding the second scenario, in which the data are simulated from the GCER model, it can be observed that the best-performing model corresponds to the data-generating model with a uniform prior. In this case, the performance of the CER model differs substantially from that of the GCER model, confirming its structural rigidity and limited ability to adapt to more complex scenarios. It can also be observed that the use of a more flexible prior distribution to model the concentration parameter of the CER model allows the structure of the data to be captured more effectively.

7. Analysis of the Human Brain Networks Data

The theoretical results presented in the previous chapters were applied to the ‘Human Brain Networks’ dataset. This dataset analyzes the brain connectivity of thirty individuals, each of whom underwent up to a maximum of ten measurements through diffusion magnetic resonance imaging (dMRI) scans, for a total of $n = 266$ observations.

Brain connectivity, particularly structural connectivity, provides information on the structure of physical connections among different brain regions. Analyzing structural connectivity is essential for identifying abnormalities that may indicate the presence of neurological and psychiatric disorders. Each observation in the dataset, as shown in Figure 7.1, corresponds to an undirected labeled binary graph with 48 nodes. Each node represents a brain region, while the binary edges indicate whether or not a connection is present between different areas of the brain.

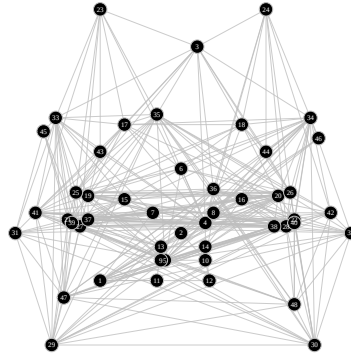


Figure 7.1: i -th observation of the Human Brain Networks dataset

The aim of the analysis is to apply the Bayesian inference framework described previously in order to estimate, for each subject, the location parameter $\mathcal{G}^m$ and the scale parameter α . To estimate the parameters of interest, the hierarchical model described in Section 4.1

was considered. Under the assumption that the scale parameter α is constant, this model exploits the conjugacy property of the GCER prior distribution defined for the parameter $\mathcal{G}^m$ with respect to the CER model. This property provides an important computational advantage, since it eliminates the need to resort to simulation-based algorithmic methods. For each patient, the scale parameter α was estimated using the “Empirical Bayes” approach described in Chapter 6, and the results obtained are reported in Table 7.1. The results show that the observations appear to be concentrated around their respective location parameter $\mathcal{G}^m$, since the values of α estimated for each patient are close to the lower bound of its domain.

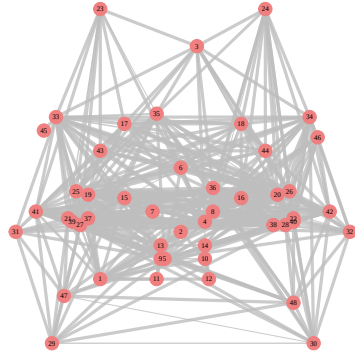
Patient 1	Patient 2	Patient 3	Patient 4	Patient 5	Patient 6
0.0621	0.0662	0.0542	0.0702	0.0596	0.0686
Patient 7	Patient 8	Patient 9	Patient 10	Patient 11	Patient 12
0.0819	0.0799	0.0726	0.0731	0.0605	0.0493
Patient 13	Patient 14	Patient 15	Patient 16	Patient 17	Patient 18
0.0769	0.0726	0.0745	0.0727	0.0723	0.0661
Patient 19	Patient 20	Patient 21	Patient 22	Patient 23	Patient 24
0.0712	0.0603	0.0563	0.0503	0.0502	0.0661
Patient 25	Patient 26	Patient 27	Patient 28	Patient 29	Patient 30
0.0969	0.0601	0.0664	0.0639	0.06280	0.0695

Table 7.1: Values of α estimated for each patient using the “Empirical Bayes” approach with $\nu_0 = 1$ for the CER model with a GCER prior

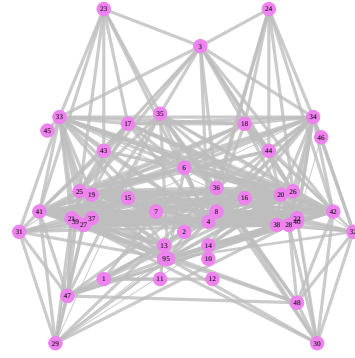
Regarding the estimation of the location parameter $\mathcal{G}^m$, it was obtained by sampling, for each patient, from the GCER posterior distribution (4.3).

Figure 7.2 shows examples of the estimated location matrices $\mathcal{G}^m$ for six of the patients under study. Each node represents a different region of the brain, while the edges indicate the presence of a connection between different brain regions. In these representations, the thickness of the edges depends on the magnitude of the probability of observing a connection between pairs of nodes. The thicker the edges, the higher the probability associated with them. Conversely, the thinner the edges, the lower the probability of observing a connection between distinct pairs of nodes. The uncertainty associated with each connection makes it possible to highlight potential structural differences among the configurations of the $\mathcal{G}^m$

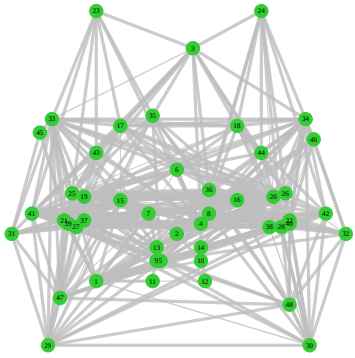
matrices and, consequently, among the brain systems of the patients.



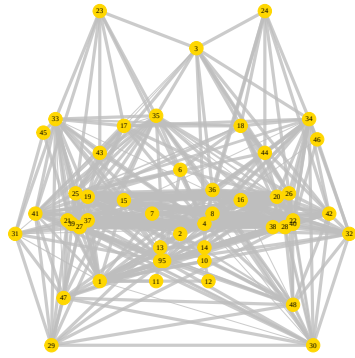
(a) Patient 2



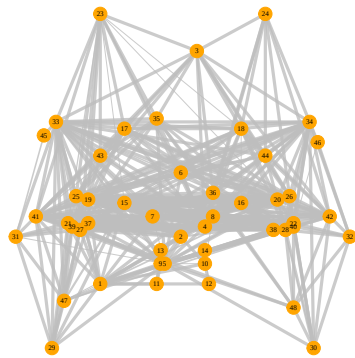
(b) Patient 5



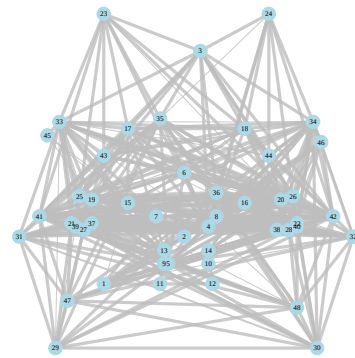
(c) Patient 10



(d) Patient 15



(e) Patient 20



(f) Patient 30

Figure 7.2: Estimates of $\mathcal{G}^m$ for selected patients.

Similar results for the estimates of the parameters α and $\mathcal{G}^m$ were obtained by applying the methodologies described in Section 4.2, in the case where α is considered a random variable. Tables 7.2 and 7.3 show that the estimates of the scale parameter obtained using the “Empirical Bayes” approach are consistent with those obtained using a fully Bayesian approach. In the scenario where the likelihood function and the prior distribution for the location parameter $\mathcal{G}^m$ share the same scale parameter α , the joint distribution of the parameters $(\mathcal{G}^m, \alpha)$ was estimated by implementing a Gibbs sampling algorithm combined with a Metropolis–Hastings Random Walk, with a total of 10000 iterations, of which 1000 were used as burn-in. Under the assumption of independence between the likelihood function and the prior distribution for $\mathcal{G}^m$, a standard Gibbs sampling algorithm was instead implemented with the same number of iterations and burn-in iterations. In both cases, $a_0 = 2$ and $b_0 = 3$ were chosen as the hyperparameters of the Beta prior distribution for α .

Patient	α Value	Patient	α Value
1	0.0623	2	0.0663
3	0.0548	4	0.0706
5	0.0598	6	0.0692
7	0.0827	8	0.0806
9	0.0732	10	0.0736
11	0.0608	12	0.0496
13	0.0772	14	0.0732
15	0.0748	16	0.0733
17	0.0727	18	0.0664
19	0.0714	20	0.0606
21	0.0566	22	0.0505
23	0.0505	24	0.0663
25	0.0983	26	0.0603
27	0.0668	28	0.0641
29	0.0633	30	0.0700

Patient	α Value	Patient	α Value
1	0.055	2	0.057
3	0.0530	4	0.0710
5	0.0600	6	0.0703
7	0.0840	8	0.0800
9	0.0711	10	0.0730
11	0.0600	12	0.0501
13	0.0802	14	0.0740
15	0.0750	16	0.0740
17	0.0730	18	0.0670
19	0.0720	20	0.0610
21	0.0577	22	0.0510
23	0.0542	24	0.0702
25	0.0990	26	0.0600
27	0.0670	28	0.0645
29	0.0632	30	0.0731

Table 7.2: Estimated values of α , dependent-prior scenario with $\nu_0 = 1$.

Table 7.3: Estimated values of α , independent-prior scenario with $\alpha_0 = 0.1$.

Figure 7.3 reports the diagnostics related to the estimation of α_i , where α_i denotes the estimate of the scale parameter for the i -th subject obtained by applying the simulation-based approaches described previously. By examining the trace plots, it can be observed that both algorithms appear to have reached convergence, since the trajectories of the estimates fluctuate stably around a mean value.

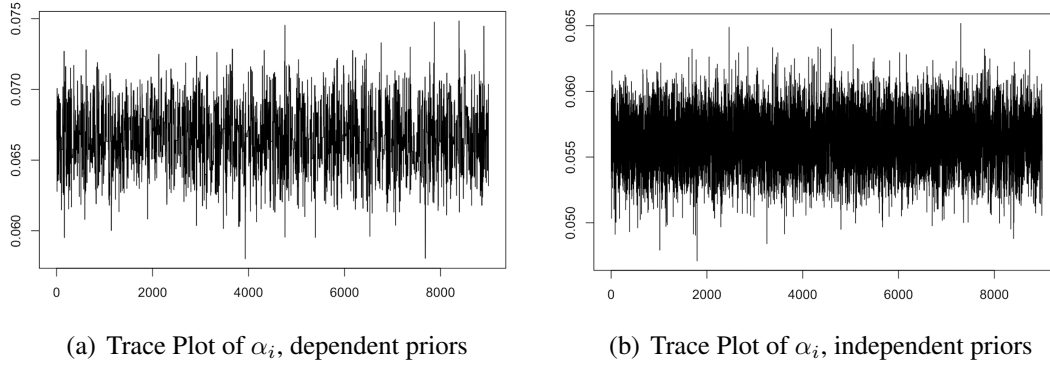


Figure 7.3: Trace plot

The results obtained from the estimation of the parameters $(\mathcal{G}^m, \alpha)$, by applying the CER model with a GCER prior distribution for $\mathcal{G}^m$, were compared with those obtained from the CER model with a prior distribution of the same type, in order to assess whether the two approaches exhibit any differences.

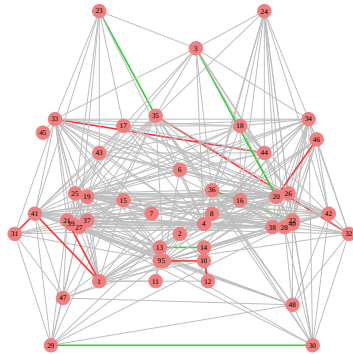
Also in this second scenario, the scale parameter α for each patient was estimated using the “Empirical Bayes” approach. Regarding the location parameter $\mathcal{G}^m$, a Gibbs sampling algorithm was implemented with 10000 iterations, of which 1000 were used as burn-in.

Table 7.4 reports the estimated values of α for each patient. Also in this case, it can be observed that the values of the scale parameter are very similar to each other. Comparing these values with those reported in Table 7.4, it can be observed that the parameters estimated using the CER distribution as the prior distribution for $\mathcal{G}^m$ have a greater magnitude.

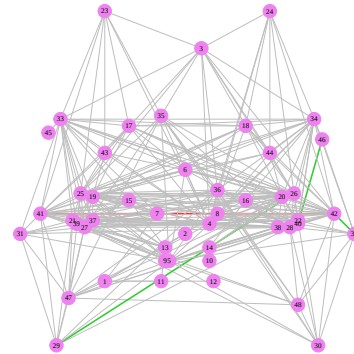
Patient 1	Patient 2	Patient 3	Patient 4	Patient 5	Patient 6
0.0998	0.0965	0.0936	0.1146	0.0962	0.1148
Patient 7	Patient 8	Patient 9	Patient 10	Patient 11	Patient 12
0.1122	0.1216	0.1058	0.1064	0.0946	0.1087
Patient 13	Patient 14	Patient 15	Patient 16	Patient 17	Patient 18
0.1065	0.1143	0.1083	0.1102	0.112	0.1065
Patient 19	Patient 20	Patient 21	Patient 22	Patient 23	Patient 24
0.1006	0.0927	0.1008	0.1183	0.0892	0.1050
Patient 25	Patient 26	Patient 27	Patient 28	Patient 29	Patient 30
0.1419	0.0948	0.0998	0.0975	0.01209	0.1142

Table 7.4: Values of α estimated for each patient using the “Empirical Bayes” approach, CER model with a CER prior

Figure 7.4 shows the location parameters $\mathcal{G}^m$ for six different patients. Each graph highlights the connections estimated using the two modeling scenarios described above. The green lines represent the connections estimated exclusively by the CER model with a CER prior distribution, while the red lines represent the connections estimated by the same CER model but with a GCER prior distribution. The gray lines, instead, identify the connections common to both approaches. Overall, the location matrices estimated by the two methods are very similar, showing no substantial differences between the two estimation approaches.



(a) Patient 2



(b) Patient 5

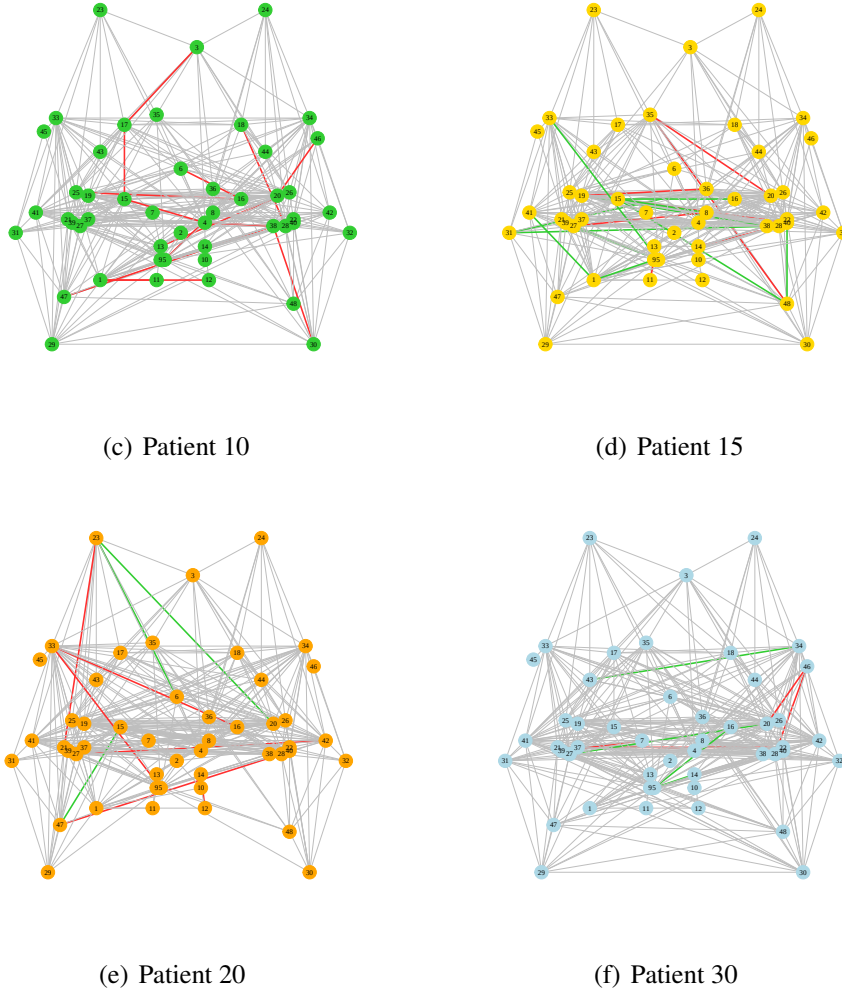


Figure 7.4: Estimates of $\mathcal{G}^m$ for six selected patients.

The second part of the analysis focuses on presenting an application example of the methodologies described in the previous chapters, particularly discriminant analysis. The main objective is to evaluate the ability of the CER model with a GCER prior distribution on the parameter $\mathcal{G}^m$ to classify new patients based on their structural characteristics. The following study includes 30 distinct subjects, each of whom underwent multiple screening sessions. The aim of the analysis is to determine the group membership of each subject undergoing a new screening based on their brain characteristics. Under the assumption of heteroscedastic observations, for each patient, once the location parameter $\mathcal{G}^m$ and the scale parameter α were estimated, the discriminant ability of the model was evaluated by applying the classification rule (5.2). For this purpose, a k -fold cross-validation with 10 folds was performed in order to obtain a robust estimate of the model performance. Accuracy was used as the main metric to evaluate the classification performance of the model. This quantity measures

the percentage of correct classifications out of the total number of observations and is useful because it provides a summary measure of the model's ability to distinguish among the subjects' classes. High accuracy values indicate good classification performance, whereas lower values indicate that the model has difficulty accurately discriminating new observations.

The CER model with a GCER prior distribution shows excellent discriminant ability, achieving a cross-validated accuracy of 99%. Based on the estimates of the scale parameters α reported in Table 7.1, the observations appear to be highly concentrated around their respective location parameters, thus identifying groups that are highly homogeneous internally and characterized by low variability. This contributes to the good classification performance of the model.

Figure 7.5 reports the results obtained from the discriminant analysis; each symbol represents a distinct category of patients, while the blue squares represent the subjects who were not correctly classified into their respective groups. As can be observed, out of 266 observations, only two were misclassified, specifically two observations belonging to the patient classes with identifiers 0025433 and 0025451. In the figure, the subjects are represented according to two quantities: 'Average Path Length' and 'Clustering Coefficient'. The first metric measures the average distance between pairs of nodes in a network and is defined as follows:

$$L = \frac{1}{N(N-1)} \sum_{i \neq j} d(i, j),$$

where N is the number of nodes characterizing the network and $d(i, j)$ represents the shortest distance between node i and node j . This quantity makes it possible to characterize the efficiency with which information propagates through the network. High values of 'Average Path Length' indicate that many steps in terms of edges are required to move from one node to another, whereas low values indicate that the network has good connectivity.

The 'Clustering Coefficient', instead, measures the local connectivity of the network, that is, whether there are subgroups of closely connected nodes. This quantity is defined as follows:

$$C = \frac{1}{N} \sum_{i=1}^N C_i,$$

where

$$C_i = \frac{2E_i}{k_i(k_i - 1)}.$$

The quantity E_i corresponds to the number of edges among the neighbors of node i , where neighbors are defined as all nodes that share an edge with node i . The coefficient k_i , instead, denotes the degree of node i , that is, the number of neighboring nodes. High values of the ‘Clustering Coefficient’ indicate that the network contains local subgroups, whereas low values indicate that the network has a random structure.

The purpose of this representation is to highlight whether some patients exhibit similar neurological patterns, in order to identify potential brain abnormalities and, consequently, possible pathologies. Figure 7.5 shows that the subjects tend to be randomly distributed within the space defined by the two metrics, indicating that the patients under study exhibit a broadly similar neurological configuration.

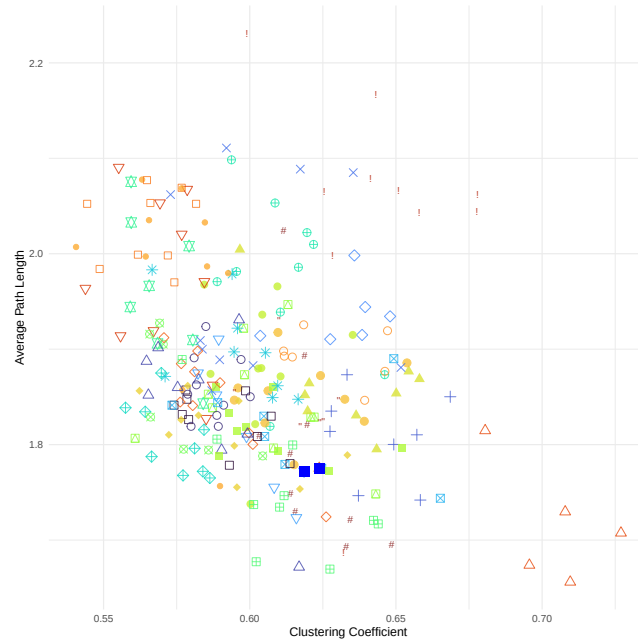


Figure 7.5: Results of the cross-validated discriminant analysis for the CER model with a GCER prior

The results obtained by applying the CER model with a GCER prior distribution were compared with those obtained by applying the same model but with a CER prior distribution. Also in this scenario, the model shows excellent discriminant ability, achieving an accuracy of 98%.

Figure 7.6 shows that, out of 266 observations, only four were misclassified.



Figure 7.6: Results of the cross-validated discriminant analysis for the CER model with a CER prior

Overall, both specifications show excellent ability to classify patients based on their structural characteristics. The use of a GCER prior distribution leads to an improvement in terms of accuracy, although the improvement is minimal. This application example does not reveal a substantial difference for the CER model between the use of a CER prior and a GCER prior, since the observations considered exhibit low variability and are highly concentrated around their respective location measures, ensuring excellent classification performance in both scenarios. As shown in the simulation study, the introduction of a more flexible prior distribution improves the model fit, although in this application example such an improvement is only marginally evident.

Conclusions

In this thesis, a new extension of the CER model, called GCER, was introduced. In particular, it was shown that this model provides greater flexibility than the baseline model, since the location measure parameterizing the new distribution is no longer a binary graph but a weighted graph.

In addition to the theoretical formulation of the new probabilistic model, its properties and the relationship between the baseline CER model and its generalization were presented.

The study highlighted an important property of the GCER model concerning the inferential estimation of the location parameter, namely its conjugacy with respect to the CER model. This property provides a significant computational advantage in the estimation of the location parameter under the assumption that the scale parameter is constant.

The simulation study confirmed that the use of the GCER model, both as the model from which the sample is generated and as the prior distribution for the location parameter of the CER model, leads to an improvement in terms of goodness of fit to the data. This result highlights how more flexible models can better represent the data compared with more rigid models, such as the CER model.

In the application study, the CER model was assumed to be the data-generating model, and the impact of adopting a more flexible prior distribution for the location parameter was evaluated. The inferential techniques applied to the real ‘Human Brain Networks’ dataset for the estimation of the location and scale parameters did not reveal substantial differences between the results obtained using the CER and GCER specifications to define the prior distribution of the location parameter of the CER model.

The discriminant analysis showed that both model specifications exhibit excellent classification performance, although the introduction of a GCER prior leads to a slight improvement in terms of classification performance.

The application study was limited to comparing the CER model using two different prior specifications for the location parameter, leaving the use of the GCER model as the data-generating model for future investigation.

As shown in Chapter 3, using the GCER model to model the data has the disadvantage of being computationally expensive when the number of nodes characterizing the graph is large. Indeed, the functional form of the GCER model is analytically complex, making it diffi-

cult, when performing inference on the parameters characterizing the distribution, to obtain a known or easily tractable functional form for the posterior distribution.

It is also necessary to highlight an important aspect concerning the structure of the GCER probabilistic model, namely its identifiability. A probabilistic model is identifiable if different combinations of parameters generate distinct probability distributions. In the case of the GCER model, identifiability issues may arise because different combinations of parameters can generate the same probability distribution. This may represent a problem during parameter estimation, since the parameters cannot be uniquely estimated. One possible solution to this limitation is to fix one component of the location parameter W . In this way, the remaining components of W and the parameter α can be uniquely estimated.

Bibliography

- Barile, F., Lunagómez, S. & Nipoti, B. (2024). Bayesian nonparametric modeling of heterogeneous populations of networks. *ArXiv* .
- Christensen, R., Johnson, W., Branscum, A. & Hanson, T. (2010). *Bayesian Ideas and Data Analysis*. Chapman & Hall.
- Dubey, P. & Müller, H.-G. (2019). Fréchet analysis of variance for random objects. *Biometrika* .
- Geisser, S. & Eddy, W. F. (1979). A predictive approach to model selection. *Journal of the American Statistical Association* **74**, 153–160.
- Heard, N. (2021). *An Introduction to Bayesian Inference, Methods and Computation*. Springer.
- Hoff, P. D. (2009). *A First Course in Bayesian Statistical Methods*. Springer.
- Lunagómez, S., Olhede, S. C. & Wolfe, P. J. (2021). Modeling network populations via graph distances. *Journal of the American Statistical Association* **116**, 2023–2040.
- Marin, J.-M. & Robert, C. P. (2014). *Bayesian Essentials with R*. Springer.
- Roberts, G. O. & Rosenthal, J. S. (2001). Optimal scaling for various metropolis-hastings algorithms. *Statistical science* **16**, 351–367.